

ZUBAIR MOSIMANEGAPE JALAL

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PROFESSIONAL SUMMARY

MSc Data Science candidate (on track for Distinction) with a BSc in Mathematics and Economics and two years' applied data-science experience, building end-to-end machine learning on multimodal data — from problem definition through modelling to deployment and monitoring. Delivers deployed fault-detection models and BI in industry, alongside an MSc dissertation predicting Alzheimer's conversion from multimodal health data with SHAP explainability. Complemented by financial-services simulations (Citi, Lloyds, TATA) across regulated risk analytics and responsible AI. Advanced Python, SQL and R, with a strong record of communicating findings to non-technical stakeholders.

EDUCATION

MSc Data Science with Industry — University of Leicester Sep 2025 – Sep 2026

On track for Distinction · Botswana Top Achiever Scholarship (full government scholarship, nation's highest performers)

- Ranked #1 of cohort in Excel for Data Science and #3 in Statistics for Data Science.

BSc (Hons) Mathematics and Economics — University of Liverpool Sep 2022 – Jun 2025

Key modules: Financial Mathematics, Financial & Actuarial Modelling, Securities Markets, Financial Reporting, Linear Statistical Models.

A-Levels — Livingstone Kolobeng College, Botswana 2016 – 2021

Mathematics (A), Economics (A), Physics (A), Business (B)

WORK EXPERIENCE

Junior Data Scientist — The Digital Glow (Remote, Contract) 2024 – Present

- Built a two-stage fault-detection system in Python — an XGBoost baseline on engineered telemetry features and an autoencoder for anomaly detection — across 200 live solar systems, raising fault-catch rate from 78% to 86% and cutting average downtime per fault from 3.5 to 2 hours.
- Engineered SQL pipelines aggregating multi-source telemetry and CRM data, and applied SHAP to make fault drivers interpretable for the operations team.
- Built Power BI dashboards for BAU monitoring of production, system health and commercial KPIs, with ongoing monitoring of deployed models for performance drift.

Match Official — University of Liverpool Campus Football Oct 2023 – May 2025

- Tracked disciplinary trends across a 16-team league and presented PowerPoint analytics, negotiating logistics using availability data to drive a 20% rise in fixture completion and a 10% fall in formal grievances.

Academic Mentor & Tutor — Livingstone Kolobeng College Mar 2019 – Jul 2021

- Diagnosed subject-specific weaknesses for four students and built data-driven learning roadmaps and parent reports, lifting average assessment scores 17%.

PROJECTS

MCI-to-Alzheimer's Conversion Prediction — Python, Multimodal ML (MSc dissertation) Jun 2026

- Building a clinical decision-support system on the OASIS-3 multimodal dataset (1,378 participants; 277 MCI patients), integrating 10+ neuroimaging, cognitive, clinical and biomarker files into a unified longitudinal pipeline.
- Developed and evaluated supervised, deep-learning and survival models (LightGBM, XGBoost, CatBoost, Random Forest, LSTM, Cox proportional hazards) with leakage-safe patient-level GroupKFold cross-validation and Optuna tuning.
- Applied SHAP explainability to surface key clinical drivers and deployed a Streamlit prototype returning risk scores and PDF reports; ran a requirements survey with medical professionals.

Diabetes Risk Classification — R, Random Forest (MSc) Apr 2026

- Developed a screening classifier on a 253,000-record US health dataset, correcting severe class imbalance (84% negative) with cost-sensitive weighting and precision-recall threshold optimisation; identified a BMI–sex reversal supporting sex-specific screening thresholds.

Financial Mathematics & Stochastic Modelling — Python (MSc, Distinction) Sep – Dec 2024

- Led a four-person team building Monte Carlo simulations (1M+ iterations) of asset-price trajectories under geometric Brownian motion, benchmarked against historical data; produced a financial-instrument risk-profile report.

VIRTUAL WORK EXPERIENCE

Forage job simulations (self-directed): TATA AI Strategy — GenAI-assisted EDA, agentic-AI collections strategy, and a delinquency-risk framework with responsible-AI and regulatory considerations. Lloyds Banking Group Data Science — Random Forest churn model (ROC-AUC 0.82), pandas preprocessing, GridSearchCV tuning and feature-importance analysis. Citi Financial Analyst — capital-adequacy/RWA review and KPI reporting for senior management.

SKILLS

Machine Learning & AI: Supervised, unsupervised & deep learning (LSTM, autoencoders); gradient boosting (XGBoost/LightGBM/CatBoost); Random Forest; survival analysis (Cox PH); anomaly detection; imbalanced-data methods; tuning (Optuna)

Generative & Agentic AI: GenAI-assisted EDA and agentic-AI strategy design (Forage simulations)

Explainable AI: SHAP, feature-importance analysis; model interpretability for clinical/business stakeholders

Programming: Python (pandas, NumPy, scikit-learn), R (Tidyverse, Quarto), SQL, VBA, Git/GitHub

Deployment & Workflow: Model deployment & drift monitoring, reproducible data pipelines, cross-validation, agile & cross-functional teamwork

Visualisation & BI: Power BI, Tableau (LODs, dashboards), Excel (advanced — PivotTables, slicers, KPI dashboards)

ADDITIONAL

Professional footballer — Mochudi Centre Chiefs FC; Botswana U17/U21 international (professional contract at 18). Languages: English, Setswana, conversational Afrikaans & Arabic.